

FairKalah without Empty Capture

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COLLEGE

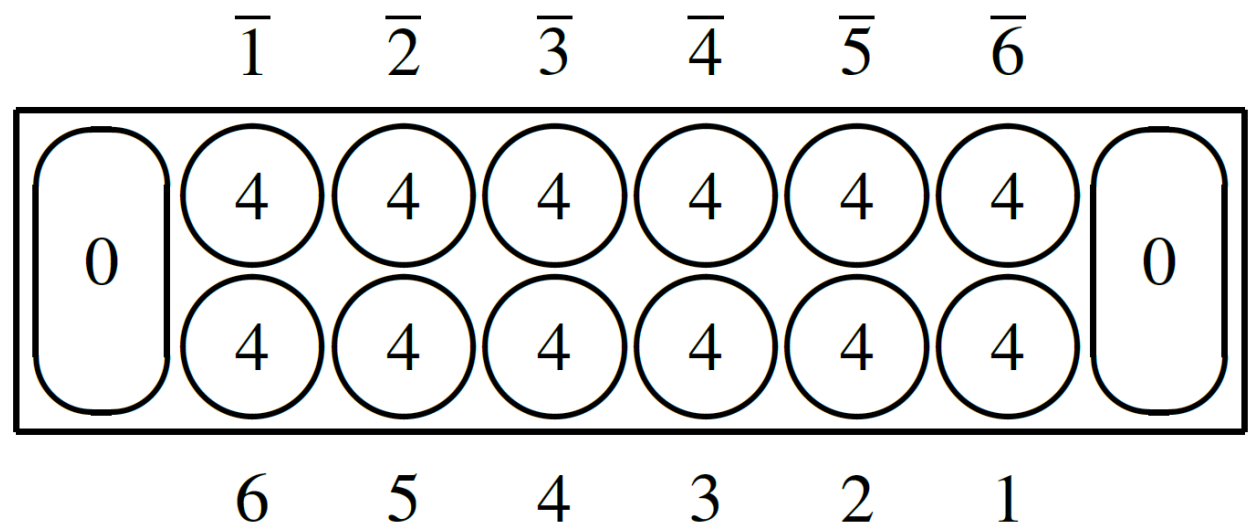


Outline

- Mancala no empty captures rules
- Mancala fair initial state (FairKalah) computation
- Fair Optimal Game Tree insights
- Diverse state optimal play modeling insights

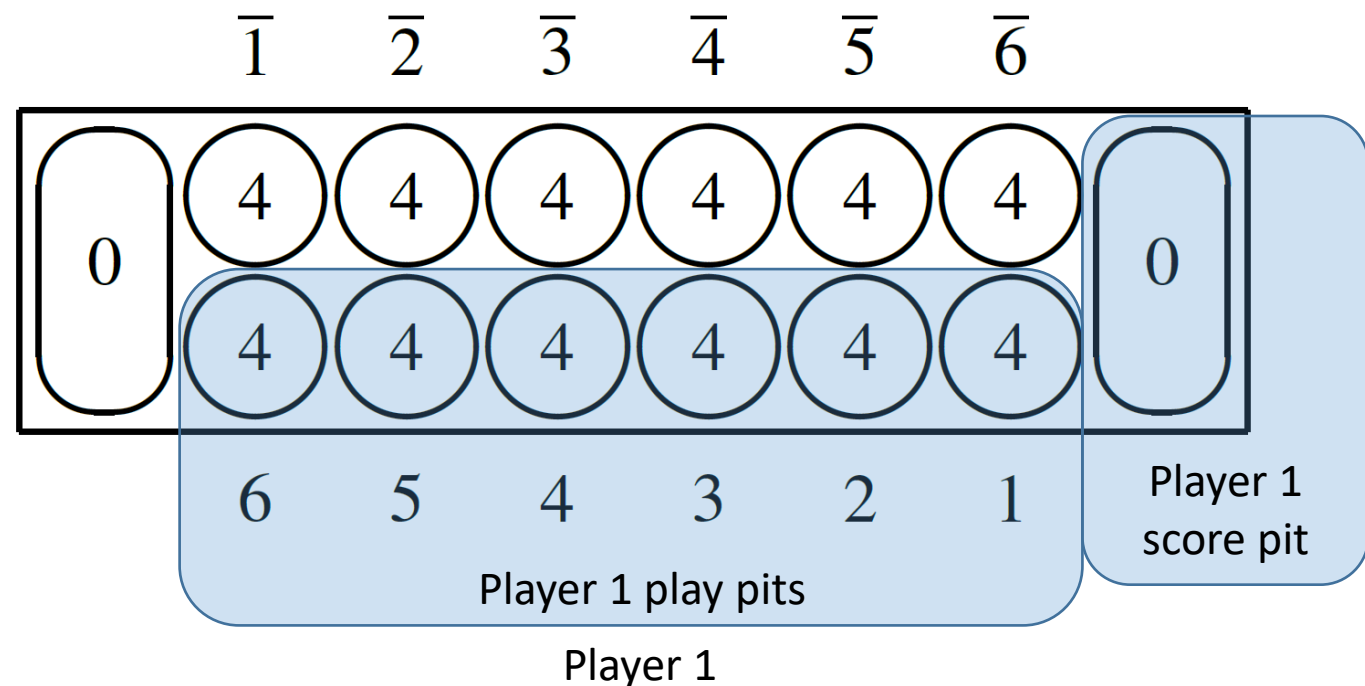
Mancala (a.k.a. Kalah) Materials

- Board with
 - 6 play pits per side for each player
 - 2 score pits, one to the right end of the board for each player
- 48 pieces initially distributed
4 per play pit in standard game



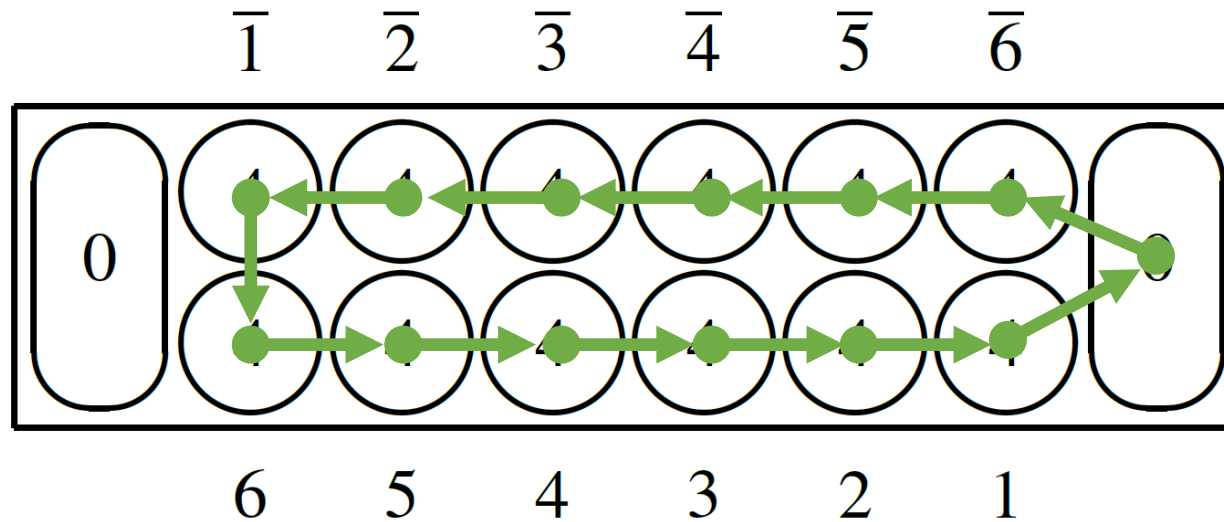
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Mancala Move

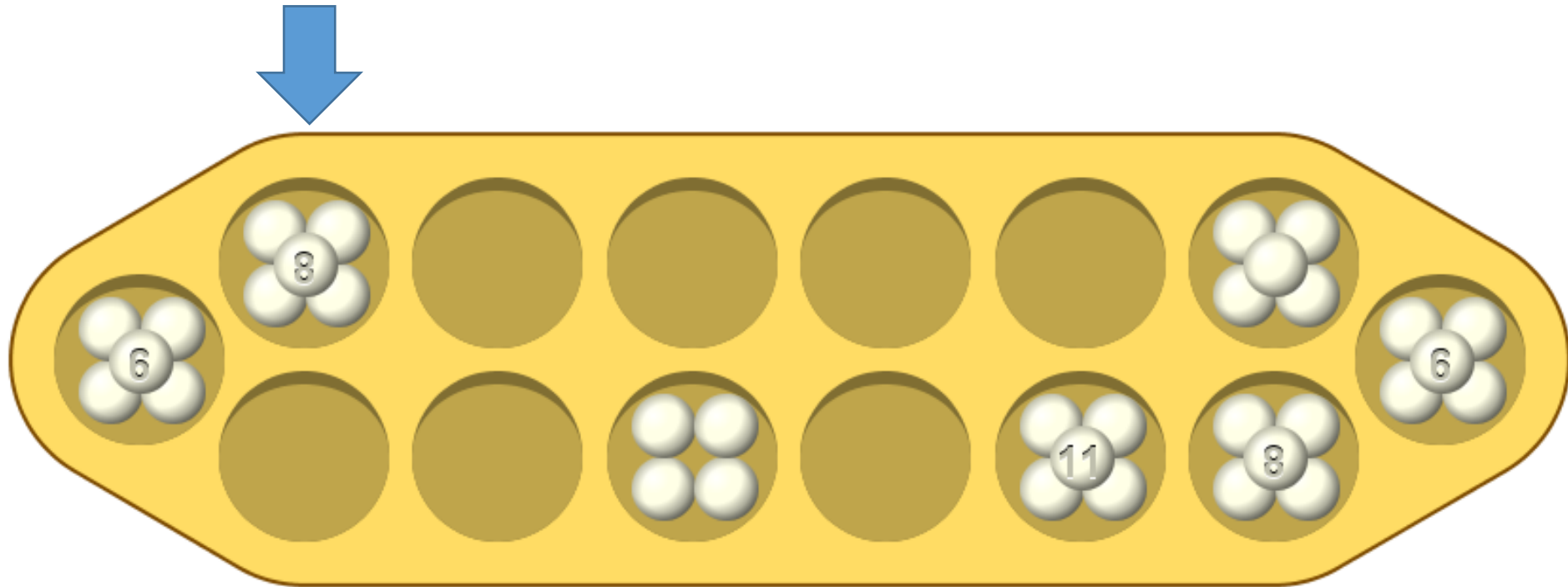
- A player's move in Mancala consists of
 - Selecting their non-empty play pit,
 - Picking up all pieces from that pit,
 - And “sowing” them counter-clockwise, one per pit, skipping the opponent's score pit.



Player 1 sowing pattern

Mancala Move Example

- Player 2 (top) plays 8 pieces from upper-leftmost play pit:

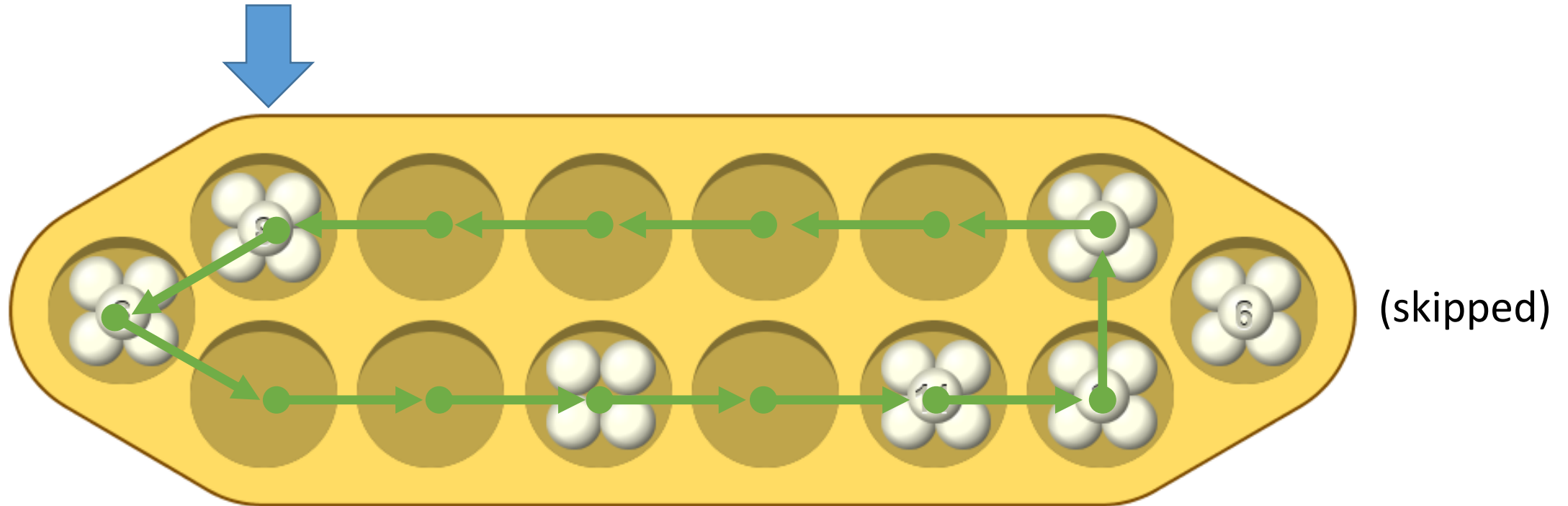


Screenshots from Ludii general game system:



Mancala Move Example

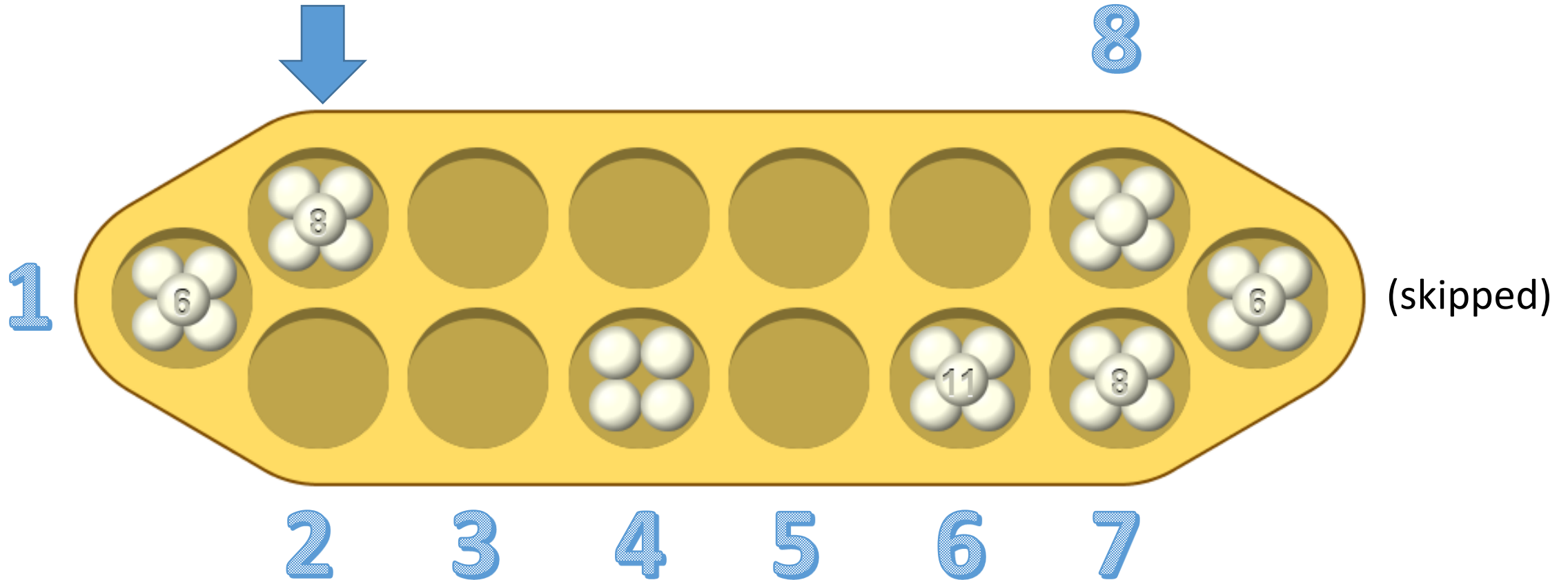
- Player 2 (top) plays 8 pieces from upper-leftmost play pit:



Player 2 sowing pattern

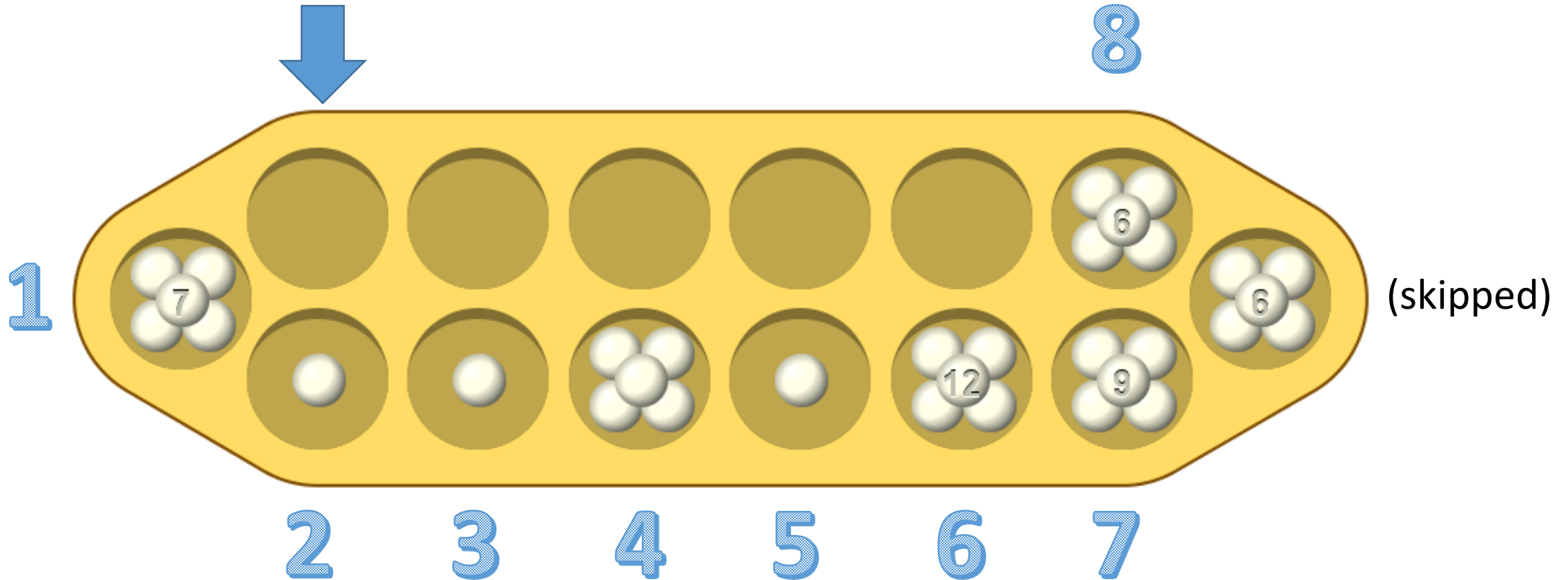
Mancala Move Example

- Player 2 (top) plays 8 pieces from upper-leftmost play pit:



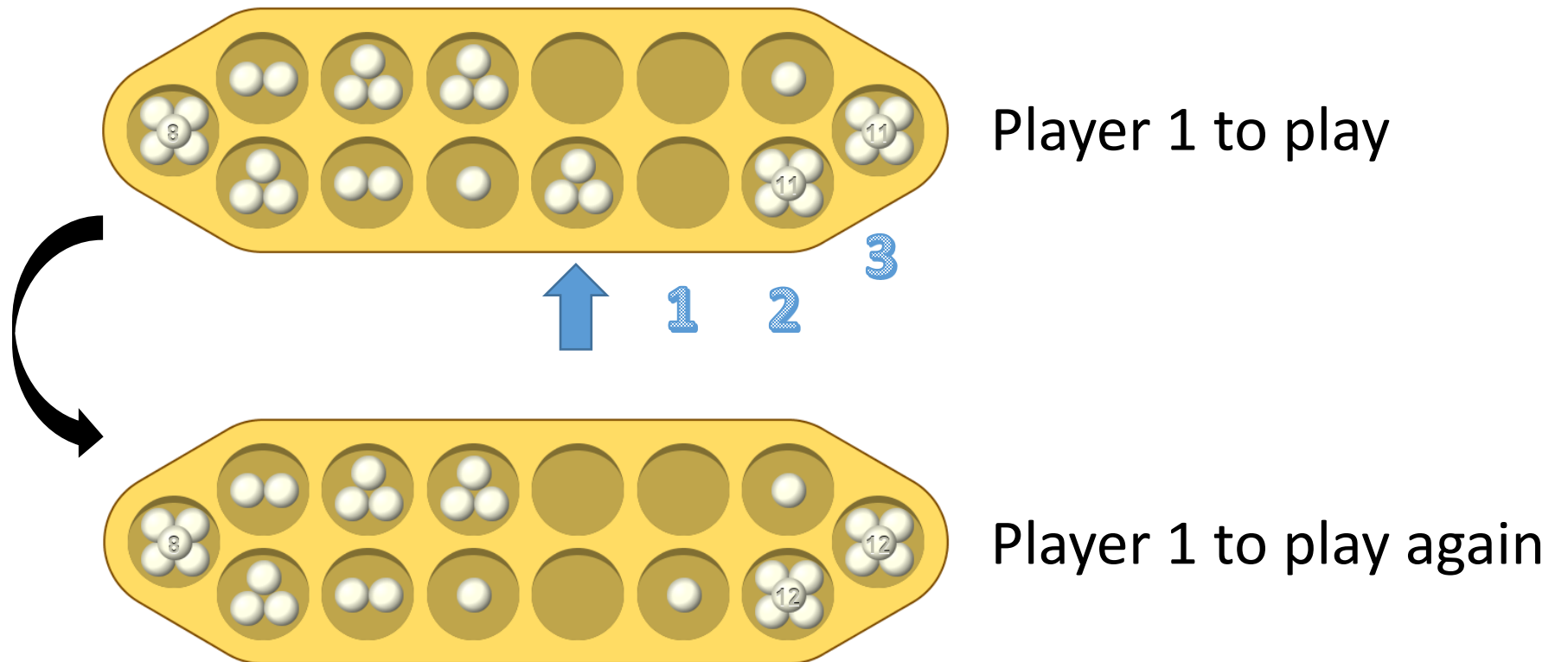
Mancala Move Example

- Player 2 (top) plays 8 pieces from upper-leftmost play pit:



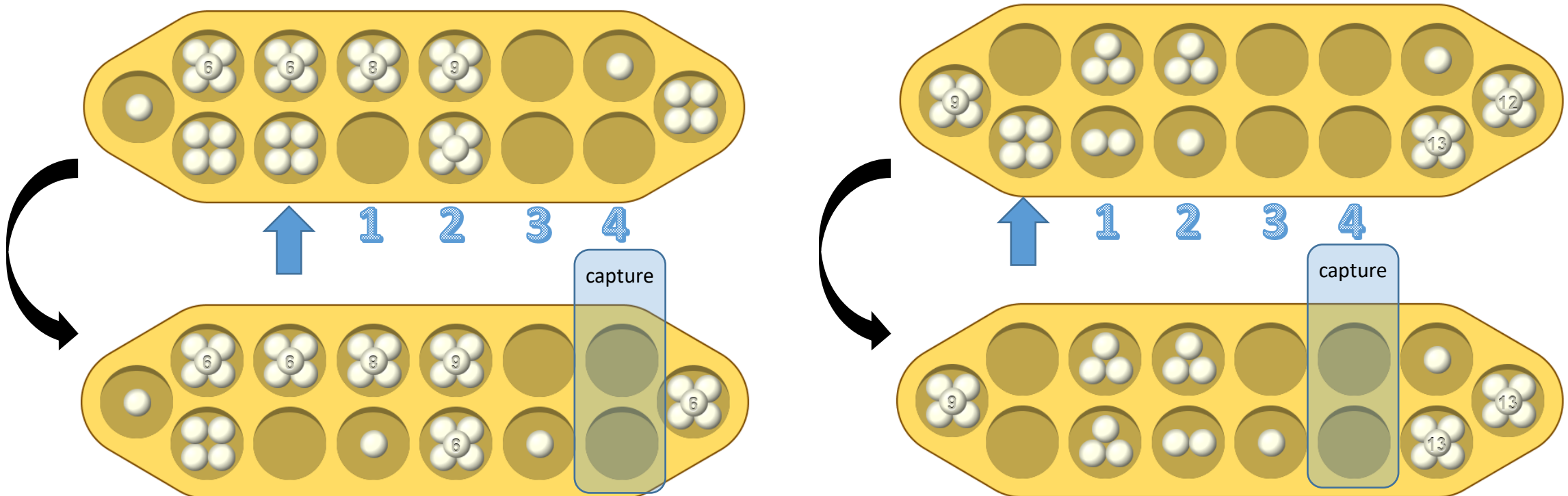
Mancala Free Move

- If your last piece sown is to your score pit, take another turn.



Previous Mancala Capture

- If your last piece sown is to an empty play pit on your side, capture that piece and any in the opponent's opposite pit (which may be empty). Captured piece(s) are placed in the player's score pit.



Empty Capture Patent Ambiguity

- US Patent #2,720,362 (“Game Counter”, W. J. Champion, Oct. 11, 1955) doesn’t explicitly list the presence of an opponent piece to capture as a condition of capture:

2. If the last game piece 22 is placed in an empty pit 21 all of the game pieces in the laterally aligned pit of the opposing player are captured. The captured pieces together with the piece making the capture are removed and placed in that player’s bowl or kalah 23. A capture ends that player’s turn.

Evidence of No Empty Capture Intention

- Circa 1963, Champion played the first Kalah AI implementation on a PDP-1 programmed by “M.I.T.’s computer wizards” (Time, “Pits and Pebbles”, June 13, 1963)
- The main site of Champion’s Kalah production was in Holbrook, Massachusetts, ~20 miles away.

Intrigued by the high number of mathematical combinations in the game, M.I.T.’s computer wizards programmed a PDP-1 computer for Kalah, which clobbered Champion consistently after the first try. “I didn’t have a chance,” he said gamely. “The computer eluded every one of my traps. It can think so many moves ahead that it’s impossible to beat.”

In his small plant at his son’s electronics firm in Holbrook, Mass., Champion turns out 24 styles of board, ranging from \$4.50 to \$25 (up to \$250 for custom-made models), and is currently negotiating with a company equipped to turn out the boards at the rate of 10,000 or more a week.

Roland Silver's Kalah AI

- The PDP-1 Kalah AI implementation was almost certainly that of Roland Silver, affiliated with MIT. This implementation explicitly disallows empty captures (DEC Interoffice Memorandum M-1098).

INTEROFFICE MEMORANDUM

M-1098

DATE March 31, 1961

SUBJECT The "Kalah" Game Playing Program

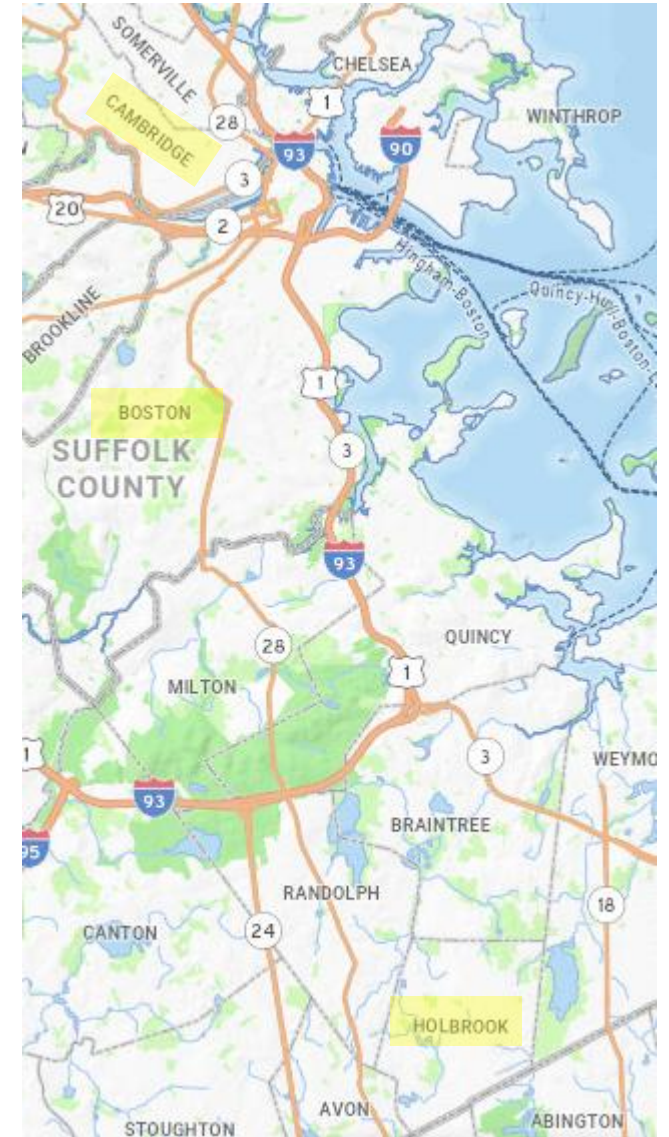
TO PDP Distribution List

FROM Roland Silver

- (c) If the last stone was cast into an empty pit on the player's side of the board, opposite a non-empty pit, then a capture results: The single stone in the previously empty pit, together with all the stones in the opposite pit, are taken up and cast into the player's kalah.

No Empty Capture Conclusion

- Given:
 - Champion likely played Silver's Kalah AI implementation in Cambridge, Massachusetts with no empty captures allowed and raised no rule objection.
 - Primary production of the Kalah Game Company was ~20 miles away in Holbrook, Massachusetts at that time.
 - Kalah was reportedly popular and played by thousands of children in nearby Boston's playground system at that time. (Time, "Pits and Pebbles", June 13, 1963).
- It is very unlikely that Roland Silver's unambiguous specification of the capture rule was incorrect. That is, Kalah was played without empty captures.

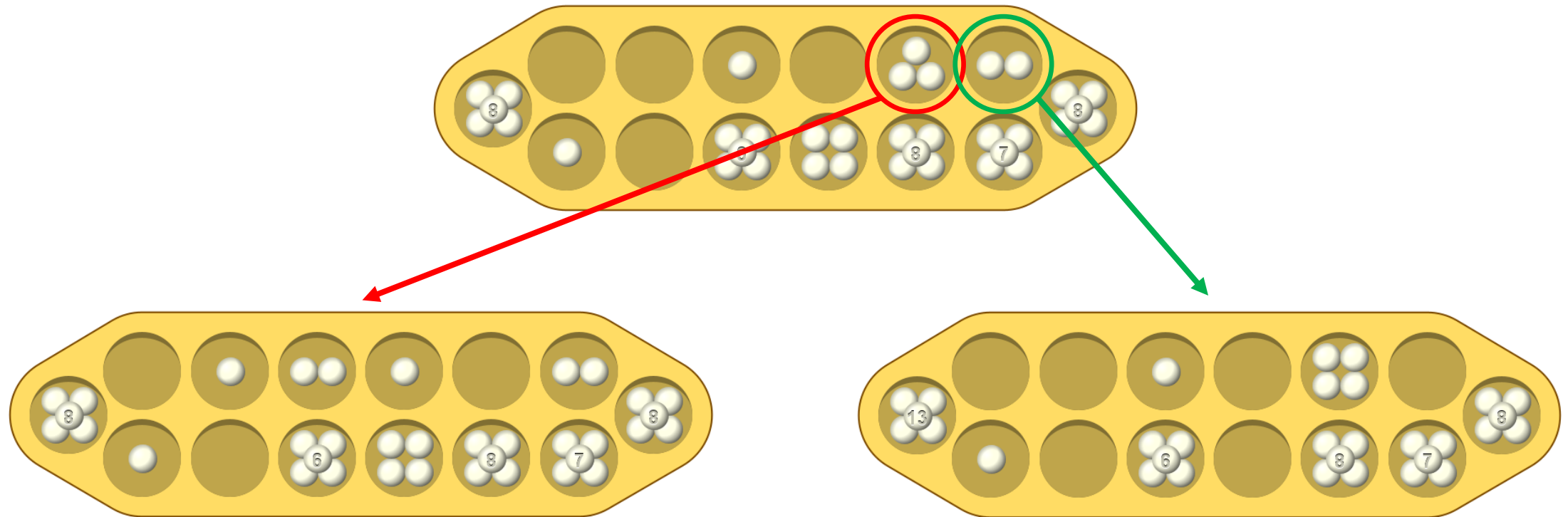


Eastern Massachusetts, USA

Mancala No Empty Capture (NEC)

- With disallowing empty captures, the player could only perform capture when last piece sown is to an empty play pit on your side, and there must be piece(s) on the opponent's opposite pit.

No Empty Capture



No Empty Capture

Capture

Mancala Game End

- “Starvation” - At the end of a turn, when no pieces remain in one player’s play pits, their opponent scores remaining play pits.
- The player that scores more pieces wins. If both players score the same number of pieces, the game is a draw (i.e. tie).

Problem: Mancala NEC is still Unfair

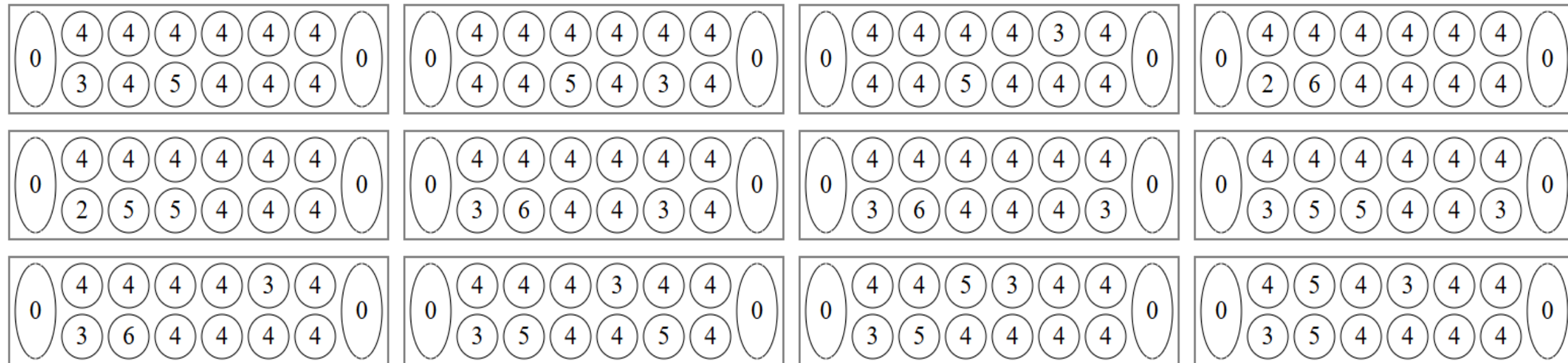
- The first of two perfect players will win by 8 points.

$m(n)$	Game Value	Perfect Play
4(1)	2 (W)	12, $\bar{3}$, 4, $\bar{121}$
4(2)	6 (W)	1, $\bar{1}$, 214, $\bar{213}$, 3121
4(3)	6 (W)	32, $\bar{3}$, 41, $\bar{4}$, 4, $\bar{13}$, 3, $\bar{12}$, 4, $\bar{1}$
4(4)	2 (W)	43, $\bar{3}$, 3, $\bar{4}$, 4, $\bar{2}$, 2, $\bar{2}$, 34, $\bar{4}$, 1, $\bar{3}$, 4, $\bar{14}$, 1213, $\bar{13}$, 12
4(5)	4 (W)	4, $\bar{4}$, 4, $\bar{2}$, 4, $\bar{3}$, 4, $\bar{2}$, 31, $\bar{21}$, 1413, $\bar{3}$, 12, $\bar{2}$, 3, $\bar{4321}$, 3, $\bar{4}$
4(6)	0 (D)	3, $\bar{4}$, 2, $\bar{1}$, 3114, $\bar{43}$, 2, $\bar{4}$, 1, $\bar{4}$, 4, $\bar{3}$, 2, $\bar{2}$, 21, $\bar{311412}$, 23, $\bar{1}$, 24, $\bar{24}$, 2, $\bar{13}$, 4, $\bar{12}$, 3, $\bar{1}$
5(1)	0 (D)	5, $\bar{5}$, 4, $\bar{4}$, 1213, $\bar{23}$, 12, $\bar{2}$, 1
5(2)	2 (W)	24, $\bar{1}$, 1, $\bar{4}$, 35, $\bar{235}$, 4, $\bar{4}$, 1213, $\bar{23}$, 12, $\bar{2}$, 1
5(3)	6 (W)	31, $\bar{2}$, 2, $\bar{53}$, 14521, $\bar{5}$, 3, $\bar{24}$, 24, $\bar{2}$, 2, $\bar{1}$, 35, $\bar{4}$, 24, $\bar{5}$, 3, $\bar{3}$, 2, $\bar{2}$, 1
5(4)	10 (W)	43, $\bar{53}$, 51, $\bar{5}$, 4, $\bar{3}$, 3, $\bar{4}$, 12, $\bar{1}$, 15, $\bar{5}$, 32, $\bar{4}$, 1, $\bar{5}$, 4, $\bar{43}$, 12, $\bar{2}$, 1, $\bar{5}$, 3, $\bar{4}$, 5, $\bar{32}$, 21
6(1)	2 (W)	12, $\bar{5}$, 6, $\bar{4}$, 5, $\bar{1213}$, 4, $\bar{12}$, 3121
6(2)	6 (W)	24, $\bar{24}$, 6, $\bar{6}$, 5, $\bar{5}$, 4, $\bar{1}$
6(3)	2 (W)	2, $\bar{32}$, 1, $\bar{46}$, 6, $\bar{4}$, 412, $\bar{231}$, 3, $\bar{6}$, 4, $\bar{2}$, 5, $\bar{5}$, 1, $\bar{5}$, 23, $\bar{6}$, 12, $\bar{5}$, 1
6(4)	8 (W)	41, $\bar{56}$, 5, $\bar{3}$, 12, $\bar{1}$, 6245, $\bar{2}$, 3, $\bar{3}$, 5, $\bar{21}$, 2, $\bar{5}$, 4, $\bar{6}$, 5, $\bar{214}$, 3, $\bar{5}$, 21, $\bar{6}$, 6, $\bar{4}$



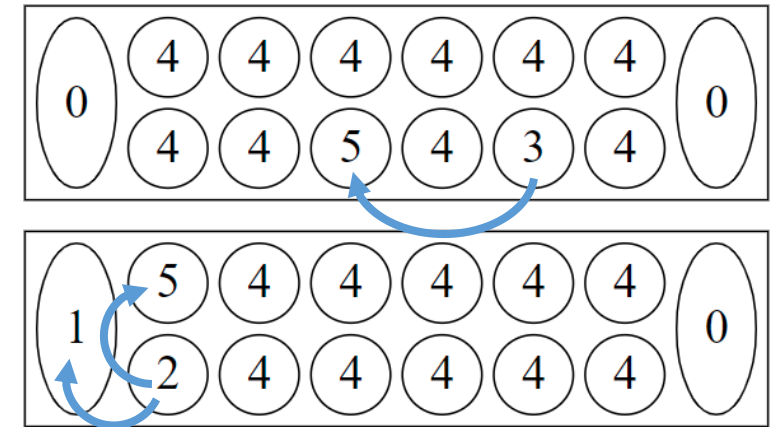
Solution: FairKalah – fair initial board states

- We have computed 323 initial states with 48 pieces arranged to be *fair*, i.e. two perfect players are proven to draw.
- This makes improvements to heuristic functions more apparent, as Mancala's unfairness obscured relative player strength.



Computation of FairKalah Boards

- For each possible board resulting from 1 or 2 pieces moved from the standard initial position, find all boards with 0 game value (1st player score – 2nd player score).
- Port of Irving's C code to Java with
 - 24-piece endgame database (1.16 GB)
 - MTD(*f*) algorithm
 - Heuristic node ordering: transposition table move, closest free moves, closest captures, closest remaining
- 3 and 320 fair boards with 1 and 2 pieces moved respectively

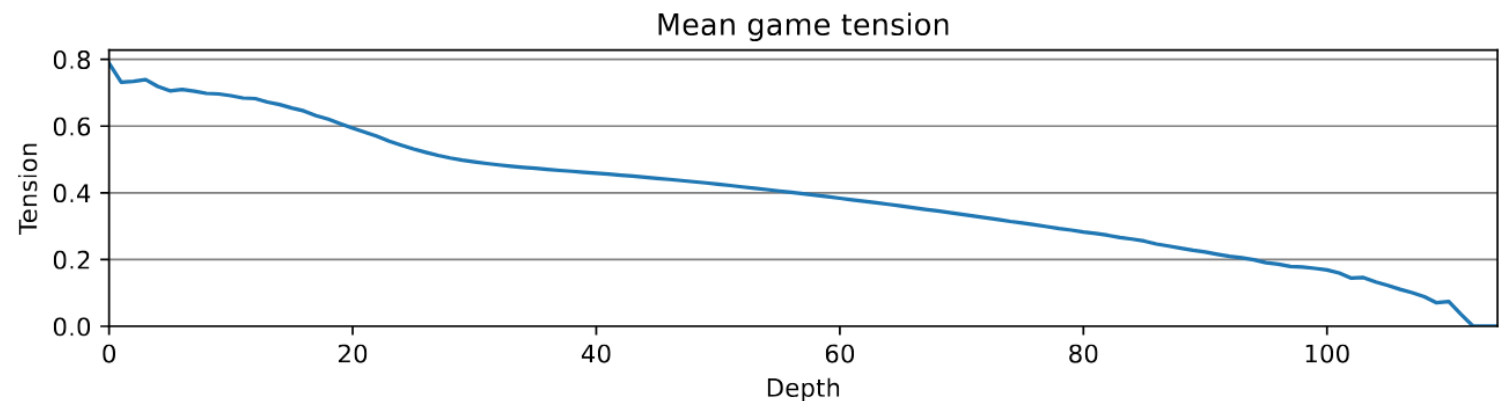
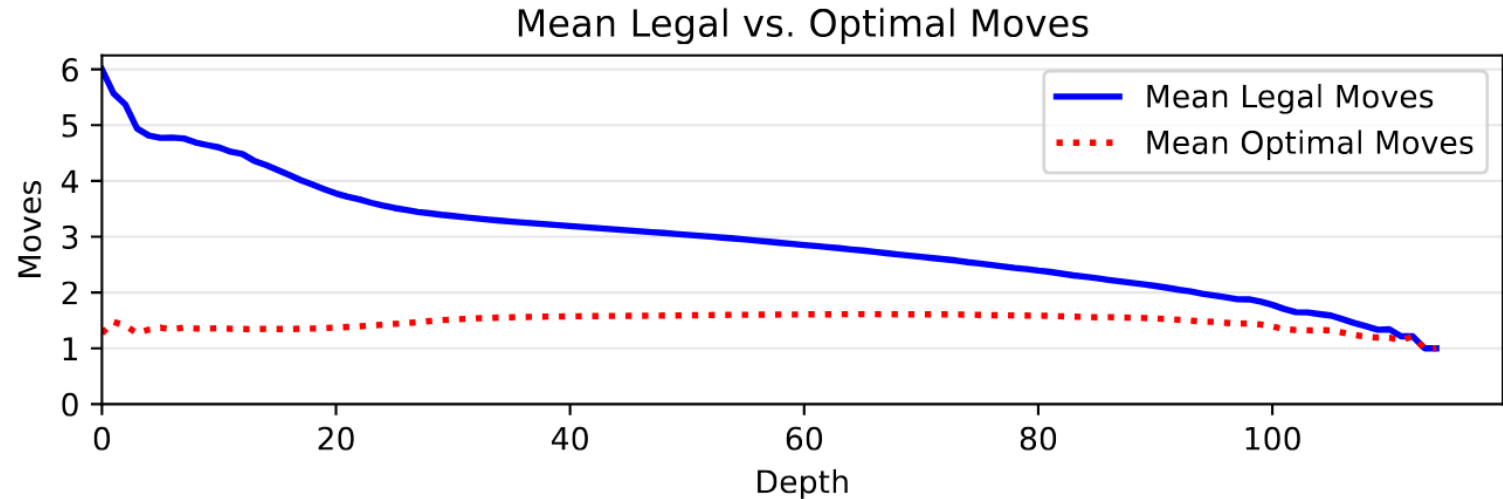


Fair Optimal Game Tree Analysis

- Fair Optimal Game Trees – subtrees of fair game tree with optimal play, i.e. all game value 0 (draw) states.
 - Pro: information about perfect play state distribution
 - Con: predictors lose performance on suboptimal game states
- MTD(f) analyses yielded 63,050,664 unique nonterminal optimal play states
 - 195,748,684 total plays and 99,308,527 optimal plays, yielding average play and optimal play branching factors of approximately 3.1046 and 1.5751, resp.
 - number of states with 1-6 optimal moves are, resp.:
36,156,747 (57.3%), 19,663,770 (31.2%), 5,446,419, 1,459,214, 298,957, and 25,557.
 - Thus, 88.5% of states have 1-2 optimal plays.

Fair Optimal Game Tree Analysis (cont.)

- With increasing depth:
 - Mean legal moves decreases from 6 to 1
 - Mean optimal moves is relatively steady
 - Thus, mean game tension decreases from .8 to 0. (.833 = unique optimal opening move)
- Conjecture: Time-management should allocate more time to opening play.



Fair Optimal Game Tree Insights

- 63,050,664 (100%) unique FOGT states:
 - 35,810,907 (56.8%) with free move(s): In 33,895,870 (**94.65%**) of these, the **closest free move is optimal**.
 - 27,239,757 (43.2%) with no free move: In 9,728,912 (35.27%) of these, capture move(s) are possible. Of these, 7,810,858 (80.28%) have an optimal capture move. Thus, **if there is no free move, in 28.31% of states, a capture move is optimal**.
 - The closer the pit, the larger the percentage of states have that pit as an optimal play.
- This supports the node ordering heuristic of Irving et al.: transposition table move, closest free moves, closest captures, closest remaining

Pit	States Optimal	Percent
1	24,617,195	39.04%
2	18,354,089	29.11%
3	15,956,613	25.30%
4	14,552,668	23.08%
5	14,035,154	22.26%
6	11,792,808	18.70%

Analysis of Diverse Play Data

- FOGT insights generalized poorly to non-fair states
- New diverse dataset:
 - First, randomly score 5 pieces
 - Breadth-First-Traversal: limited to 100 nodes per depth
 - 14,941,556 unique states, 10,925,291 after removing player and depth
- 50-50 train/test split
- Modeling packages used:
 - Linear and Logistic Regression (scikit-learn)
 - DecisionTreeRegressor/Classifier,
RandomForestRegressor/Classifier (scikit-learn)
 - CatBoostRegressor/Classifier (CatBoost)

Game Value Regression Modeling

- Regression aims to predict “game value”
 - “game_val”: The current player score minus the opponent score if the game was played to completion with perfect play
- Hyperparameters:
 - DecisionTreeRegressor: min_samples_split=500000
 - RandomForestRegressor: n_estimators=100
 - CatBoostRegressor: iterations=30, learning_rate=0.9, depth=16.

Model	MSE
Linear regression	26.48
Decision tree	8.38
Random forest	3.83
CatBoost	2.79

Optimality Classification Modeling

- Regression aims to predict individual pit optimality, e.g. whether that move is optimal
- Pits with 0 pieces pruned from analysis
- Same hyperparameters
- Base predictor:
 - Accuracy/log loss if predictor guesses that for a pit x , chance of optimality is 50%
- Log loss: stronger than accuracy

Model	6	5	4	3	2	1
Base predictor	.63	.73	.76	.79	.82	.85
Logistic regression	.80	.80	.79	.77	.78	.89
Random forest	.87	.87	.87	.87	.88	.92
CatBoost	.89	.90	.89	.89	.89	.94

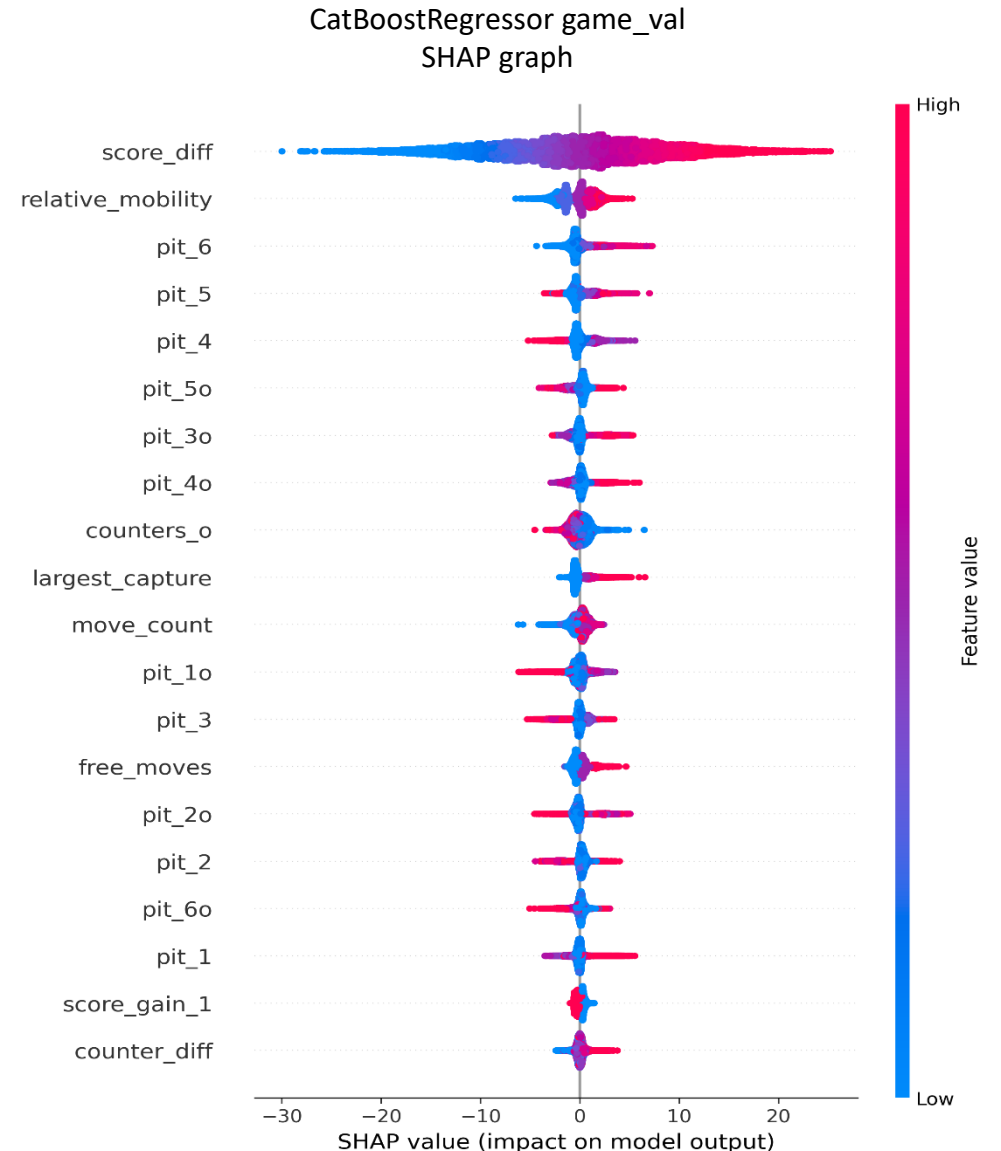
Classification Accuracies

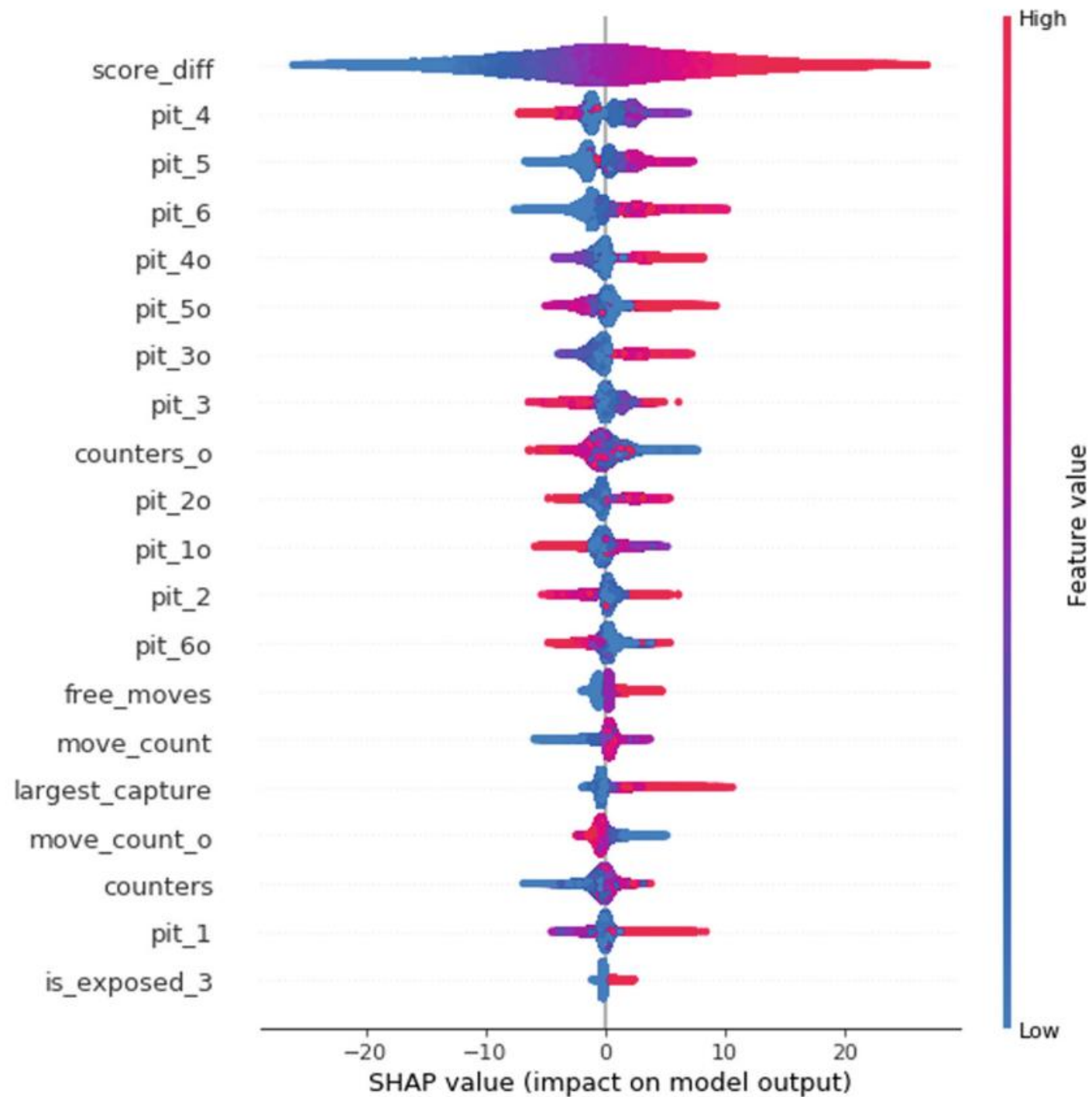
Model	6	5	4	3	2	1
Base predictor	.66	.59	.55	.51	.47	.43
Logistic regression	.43	.44	.47	.48	.46	.26
Random forest	.30	.29	.30	.29	.27	.17
CatBoost	.25	.25	.25	.25	.24	.15

Classification Log Losses

Game Value SHAP Interpretation

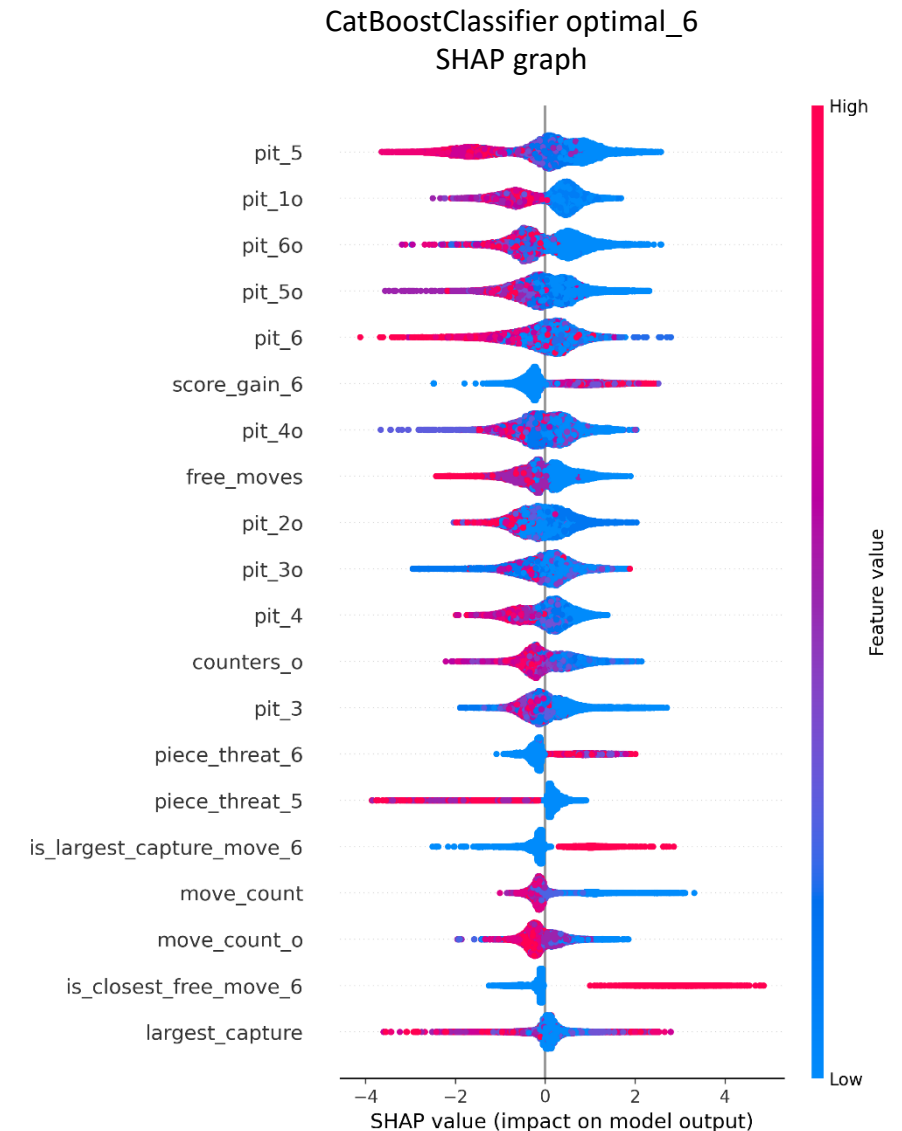
- SHapley Additive exPlanation (SHAP) values:
 - “assign each feature an importance value for a particular prediction.”
- Feature ordering is by significance, point density indicates number of occurrences
- X axis: effect on model prediction
- Interpretations:
 - Score difference highly correlated
 - No empty capture → more mobility (more moves) benefits





Pit 6 Optimality SHAP Interpretation

- Interpretations:
 - Board composition is every important, with the two closest pits and two furthest pits being most influential.
 - The more pieces in pit 6, the less likely we want to play
 - Higher score gain in pit 6 → higher chance being optimal
 - Since pit 6 is the furthest, it is rare to have it as the closest free move.
 - The least optimal pits to play from on average



Conclusions

- 323 fair starting states for Kalah (a.k.a. Mancala) moving 1-2 pieces from the standard 4-pieces-per-play-pit initial state.
- Analysis of unique fair optimal game tree states affirmed the node-ordering heuristic of Irving et al., underscoring a preference for closest free moves, closest captures, and closest plays to the scoring pit.
- Optimal play data from a diverse set of optimal and suboptimal game states yielded models with strength for predicting game values and optimal moves. Insights:
 - Decision trees ensembles perform well in modeling game value and optimality per pit
 - SHAP graphs indicate the importance of free moves, and also suggest there are overarching patterns where it is better to have more pieces in certain pits

Questions?

